

MIC Dev Recruitment Assignment

AI Mood-Based Playlist Generator Project

Project Overview

Music plays a crucial role in our daily lives, enhancing our experiences and complementing our moods. However, finding the perfect playlist that matches a specific mood or activity often requires time and effort. This project aims to solve this pain point by leveraging AI to instantly generate personalized playlists based on natural language mood descriptions.

Objective

Create an intuitive application that uses Gemini AI to interpret mood-based prompts and the Spotify API to curate personalized playlists, delivering a seamless music discovery experience. Participants may choose to develop a web application, mobile application, or both platforms.

Core Functionality

- Natural language mood/vibe input processing
- Gemini AI integration for interpreting user prompts
- Spotify API connection for playlist generation
- Display of generated playlists with track details and album art
- Basic playback controls and Spotify linking

Development Options

Option A: Web Application

Technical Requirements

- Frontend: React.js with responsive design principles
- Backend: Node.js/Express or equivalent
- Responsive UI optimized for both desktop and mobile browsers
- Clean, intuitive interface with minimal learning curve

Success Criteria

- Fully responsive web application that works across device sizes
- Complete implementation of core functionality
- Optimized user experience for browser environments

Option B: Mobile Application

Technical Requirements

- Framework: React Native or Flutter
- Native mobile UI/UX considerations
- Efficient state management
- Optimized for touch interactions and mobile screens

Success Criteria

- Native-feeling mobile application
- Complete implementation of core functionality
- Optimized user experience for mobile environments

Option C: Full-Stack Development

- Implementation of both web and mobile applications
- Consistent experience and functionality across platforms
- Optional: Shared backend or synchronization between platforms

Bonus Features (Optional)

- User authentication via Clerk
- Persistent storage of user preferences and playlist history
- Collaborative playlist generation and sharing
- Apple Music API integration (cross-service support)

Implementation Phases

Phase 1: Core Development

- Platform selection and initial setup
- Core functionality implementation
- Spotify API integration
- Gemini AI integration

Phase 2: Refinement & Enhancement

- UI/UX polish and optimization
- Implementation of selected bonus features
- Testing and bug fixes
- Performance optimization
- Deployment preparation



Submission Guidelines

GitHub Repository Requirements

- Complete source code
- [README.md](#) with:
 - Project description
 - **Screenshots/Videos**
 - Setup instructions
 - Technology stack
 - Features implemented
 - demo links

Submission link:

<https://forms.gle/WK6Abgu7rEyJEgWi6>

Deadline:

24/09/25 11:59 PM